04. Preprocessing & Feature Engineering

Understand the distinct cleaning, translation and brand-context preparation stages.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Text cleaning

- Tweets: remove RT/QT markers, URLs, redundant mentions and extra whitespace; convert emojis to text descriptors rather than deleting their sentiment signal.
- Articles: remove formatting symbols, URLs and excess whitespace; repair spaces around decimal points so numbers such as 6.81 are not split.
- Retain brand-containing mentions where relevant; indiscriminate mention removal can delete useful entities.

Language detection

The report describes langdetect (an n-gram Naive Bayes-based detector) for English, Devanagari-character detection for Hindi, and low confidence in both as a Hinglish signal. The notebook's observed labels include en, hi and hing. Language detection is imperfect on short tweets, code-switching and brand-heavy text.

Translation and transliteration

The pipeline replaces/transliterates Hindi text where possible, then translates Hindi and Hinglish into English using an unofficial free Google Translate web API via google-trans-new. Articles are split into manageable chunks; Hinglish article text is truncated to 3,500 characters before translation and translated twice in the notebook. The report notes translation is the runtime bottleneck and recommends an official paid translation API for reliable use.

Brand dictionary and regex

The team curated approximately 150 mobile brand names from public magazine sources and uses regular-expression matching rather than a learned NER model. This is a pragmatic choice for a bounded brand list, but patterns need word boundaries, aliases, case normalization and collision handling (for example, 'mi' can match ordinary text).

Brand-context segmentation

The report describes a greedy sentence-level context window: from the first occurrence of a brand to the occurrence of the next brand. The notebook uses utils.segment_by_rule() for articles and a separate tweet path. This creates brand-specific text spans for sentiment inference, but can miss long-distance references, pronouns and sentiment expressed before the brand mention.

Feature representation

- Theme classification: TF-IDF vectors feed XGBoost classifiers, separately for articles and tweets.
- Sentiment: multilingual BERT contextual representations and TF-IDF classifiers are used in different cases; lexicon and dependency-based methods were also explored.
- Headline generation: translated English article text is fed to fine-tuned T5-base.